DPriv-Bench Category 2

Corrected Counterexample Proofs (Part 1, 14 examples)

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1 Q7: Permute-and-Flip and zCDP

Original problem description

Let $\mathcal{M}$ be a Permute-and-Flip mechanism that satisfies ε-DP. Does $\mathcal{M}$ also satisfy $\varepsilon^2/8$-zero-concentrated differential privacy?

Original metadata comment

Permute-and-Flip [1] does not satisfy the bounded-range condition. Consequently, under the standard conversion, it guarantees only $\varepsilon^2/2$-DP and does not achieve $\varepsilon^2/8$-zCDP.

Corrected proof

Failure of a bounded-range sufficient condition is not itself a counterexample. We therefore instantiate the standard Permute-and-Flip mechanism. It randomly permutes the candidates and accepts candidate j with probability

$$p_j^{\text{flip}}(D) = \exp\left[\frac{\varepsilon}{2\Delta}\big(q(D,j) - q_{\max}(D)\big)\right],$$

returning the first accepted candidate. A maximum-score candidate is accepted with probability one.

Let the record domain be $\mathcal{Z} = \{(0,0),(1,-1)\}$, the candidate set be $\mathcal{Y} = \{1,2\}$, and $q((z),j) = z_j$. Take the replace-one neighbors

$$D = ((0,0)), \qquad D' = ((1,-1)).$$

Each candidate score has sensitivity $\Delta = 1$. With $r = e^{-\varepsilon}$, the two public output laws are

$$P = \left(\tfrac{1}{2}, \tfrac{1}{2}\right), \qquad Q = \left(1 - \tfrac{r}{2}, \tfrac{r}{2}\right).$$

This pair satisfies the required ε-DP premise: for example $P(2)/Q(2) = e^{\varepsilon}$ and $Q(1)/P(1) = 2 - r \le e^{\varepsilon}$, while the reverse ratios are no larger.

At order two,

$$D_2(P\|Q) = \log\left[\frac{1}{2}\left(e^{\varepsilon} + \frac{1}{2 - e^{-\varepsilon}}\right)\right].$$

For $\varepsilon = 1$ this equals

$$0.510119874355250\ldots > \frac{1}{4} = 2 \cdot \frac{\varepsilon^2}{8}.$$

Thus the finite-order requirement of $\varepsilon^2/8$-zCDP fails even though the chosen mechanism does satisfy the stated ε-DP premise.

Verdict: **No.**

2 Q20: Count-score exponential mechanism under replace-one adjacency

Original problem description

The following pseudocode is presented as an implementation of the exponential mechanism. Does it ensure ε-DP under replace-one adjacency?

Input: $D = \{z_i\}_{i=1}^n$, candidate set Y, score $q(D, y) = \sum_{i=1}^n \mathbf{1}\{z_i = y\}$, and privacy parameter ε.

Sampling rule:

$$p_D(y) = \frac{\exp(\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(\varepsilon q(D, y'))}.$$

Sample one $\hat{y}$ according to $\{p_D(y) : y \in Y\}$ and return $\hat{y}$.

Original metadata comment

The exponent in the density of the exponential mechanism is missing a factor of $1/2$. Consequently, under the replace-one neighboring relation, it satisfies only 2ε-DP.

Corrected proof

The comment states an upper bound but does not exhibit a violation of the requested bound. Let $Y = \{a, b\}$ and $t = e^{\varepsilon} > 1$. Take

$$D = (a, b), \qquad D' = (b, b), \qquad E = \{a\}.$$

These datasets are replace-one neighbors. Their event probabilities are

$$p_D(E) = \frac{1}{2}, \qquad p_{D'}(E) = \frac{1}{1 + t^2}.$$

Consequently,

$$\frac{p_D(E)}{p_{D'}(E)} = \frac{1 + t^2}{2} > t = e^{\varepsilon},$$

where strictness is equivalent to $(t - 1)^2 > 0$. This is a direct pure-DP violation.

For completeness, the actual global guarantee is 2ε-DP: under one replacement, an individual output weight and the normalizing constant can each change by a factor at most e^{ε}. The sequence $D_n = (a, b, \ldots, b)$ and $D'_n = (b, \ldots, b)$ makes the above ratio tend to $e^{2\varepsilon}$, so the worst-case constant is tight as a supremum.

Verdict: **No.**

3 Q21: Count-score exponential mechanism under add/remove adjacency

Original problem description

Use the same count score and sampling rule as in Q20,

$$p_D(y) = \frac{\exp(\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(\varepsilon q(D, y'))}.$$

Does this implementation ensure $(\varepsilon/2)$-DP under the *add/remove* neighboring relationship?

Original metadata comment

For a monotone query q, the exponential mechanism with $p_y \propto \exp(\varepsilon q/\Delta_q)$ is ε-DP [2], which is weaker than the guarantee requested in the statement, namely $(\varepsilon/2)$-DP.

Corrected proof

An ε-DP upper bound alone does not rule out a better $(\varepsilon/2)$ bound. Retain the required add/remove adjacency. With $Y = \{a, b\}$ and $t = e^{\varepsilon}$, take

$$D = (b), \qquad D' = (a, b) = D \cup \{a\}, \qquad E = \{a\}.$$

Then

$$p_D(E) = \frac{1}{1+t}, \qquad p_{D'}(E) = \frac{1}{2},$$

and hence

$$\frac{p_{D'}(E)}{p_D(E)} = \frac{1+t}{2} > \sqrt{t} = e^{\varepsilon/2},$$

because $(\sqrt{t}-1)^2 > 0$. This directly violates $(\varepsilon/2)$-DP.

Under add/remove adjacency, only the newly added candidate's weight is multiplied by t; comparing each output and the two normalizers shows that the mechanism is in fact ε-DP. Repeated additions of the same candidate show that this parameter is tight.

Verdict: **No.**

4 Q22: Doubling the exponent under add/remove adjacency

Original problem description

The mechanism receives the same inputs and count score as Q20, but samples according to

$$p_D(y) = \frac{\exp(2\varepsilon q(D, y))}{\sum_{y' \in Y} \exp(2\varepsilon q(D, y'))}.$$

Does it ensure ε-DP under the *add/remove* neighboring relationship?

Original metadata comment

For a monotone query q, the exponential mechanism with $p_y \propto \exp(\varepsilon q/\Delta_q)$ is ε-DP [2]; thus the algorithm described above satisfies 2ε-DP.

Corrected proof

Again, the stated upper bound is not by itself a lower-bound witness. Use the add/remove neighbors

and event from Q21. Writing $t = e^{\varepsilon}$ gives

$$p_D(E) = \frac{1}{1+t^2}, \qquad p_{D'}(E) = \frac{1}{2},$$

so

$$\frac{p_{D'}(E)}{p_D(E)} = \frac{1+t^2}{2} > t = e^{\varepsilon}.$$

Thus ε-DP fails on the adjacency relation specified in the question. The tight general guarantees are 2ε-DP under add/remove adjacency and 4ε-DP under replace-one adjacency.

Verdict: **No.**

5 Q42: Sheffet OLS and the full joint output

Original problem description

Is the following algorithm (ε, δ)-DP?

Input: a matrix $A \in \mathbb{R}^{n\times d}$ whose row norms are at most B, privacy parameters $\varepsilon, \delta > 0$, and output dimension r.

Set

$$w^2 = \frac{8B^2}{\varepsilon}\left(\sqrt{2r\log(8/\delta)} + 2\log(8/\delta)\right).$$

Sample $Z \sim \mathrm{Lap}(B^2/\varepsilon)$. If

$$\sigma_{\min}(A)^2 > w^2 + Z + \frac{4B^2\log(1/\delta)}{\varepsilon},$$

sample a standard Gaussian $r \times n$ matrix R and return $(RA, \text{“matrix unaltered”})$. Otherwise append $wI_{d\times d}$ to A, sample a standard Gaussian $r \times (n+d)$ matrix R, and return the resulting projection with the marker “matrix altered.”

Original metadata comment

The algorithm is adapted from Algorithm 1 of Sheffet [3]. In the PTR test, the scale of the Laplace noise Z has been reduced by a factor of four, and therefore the stated privacy guarantee no longer holds.

Corrected proof

The scale comparison does not itself prove a violation, and the public branch marker alone is insufficient for Q42. We use the joint output.

Set $B = r = d = \varepsilon = 1$ and $\delta = 10^{-6}$. Let the deterministic part of the branch threshold be

$$H = 8\left(\sqrt{2\log(8\cdot 10^6)} + 2\log(8\cdot 10^6)\right) + 4\log(10^6) = 354.68730501023583\ldots.$$

Write $k = \lfloor H - 1 \rfloor = 353$ and $\theta = H - 1 - k \in [0, 1)$. Define equal-size one-column matrices

$$A = (\underbrace{1, \ldots, 1}_{k}, \sqrt{\theta}, 0)^\top, \qquad A' = (\underbrace{1, \ldots, 1}_{k}, \sqrt{\theta}, 1)^\top.$$

Every row norm is at most one, and the matrices differ in one row. Moreover,

$$\sigma_{\min}(A)^2 = H - 1, \qquad \sigma_{\min}(A')^2 = H.$$

Let U be the public event "matrix unaltered." Because $Z \sim \mathrm{Lap}(1)$,

$$\Pr_{A}(U) = \frac{e^{-1}}{2}, \qquad \Pr_{A'}(U) = \frac{1}{2}.$$

Conditioned on U, the scalar projection $Y = RA$ is $\mathcal{N}(0, H - 1)$ under A and $\mathcal{N}(0, H)$ under A'. Consider the joint event

$$E = \{U,\ |Y| \geq \sqrt{H}\}, \qquad c = \sqrt{\frac{H}{H-1}} > 1.$$

Writing Φ for the standard normal cdf,

$$\Pr_{A'}(E) = \Phi(-1), \qquad e \Pr_{A}(E) = \Phi(-c).$$

Therefore the actual $(1, 10^{-6})$-DP excess is

$$\Pr_{A'}(E) - e \Pr_{A}(E) = \Phi(-1) - \Phi(-c) = \int_1^c \frac{e^{-x^2/2}}{\sqrt{2\pi}}\, \mathrm{d}x = 0.000341585823352\ldots > 10^{-6}.$$

This is a violation on the mechanism's complete released output.

Verdict: No.

6 Q43: Sheffet OLS and the public branch marker

Original problem description

Use exactly the matrix-projection algorithm stated in Q42. Is that algorithm $(\varepsilon/2, \delta)$-DP?

Original metadata comment

The algorithm is adapted from Algorithm 1 of Sheffet [3]. In the PTR test, the scale of the Laplace noise Z has been reduced by a factor of four, and therefore the stated privacy guarantee does not hold.

Corrected proof

Use the same parameters and neighboring matrices as in Q42. For the public event U = "matrix

unaltered,"

$$\Pr_A(U) = \frac{e^{-1}}{2}, \qquad \Pr_{A'}(U) = \frac{1}{2}.$$

If $(1/2, 10^{-6})$-DP held, these probabilities would obey the DP inequality in both directions. Instead,

$$\Pr_{A'}(U) - e^{1/2} \Pr_A(U) = \frac{1 - e^{-1/2}}{2} = 0.196734670143683\ldots > 10^{-6}.$$

Here the branch marker alone is already a public witness.

Verdict: **No.**

7 Q85: mean estimation and a finite-order RDP witness

Original problem description

Does the following algorithm satisfy $(\alpha, 0.8\varepsilon)$-RDP under add/remove adjacency for every $\alpha > 1$?

Input: $\mathcal{D} \subset [l, u]$ and $\varepsilon > 0$. Set $\Delta = u - l$, normalize $\mathcal{X} = \{(x - l)/\Delta : x \in \mathcal{D}\}$, and define

$$a = \sum_{x \in \mathcal{X}} x, \qquad b = \sum_{x \in \mathcal{X}} (1 - x).$$

Draw independent $\eta_1, \eta_2 \sim \mathrm{Lap}(1/\varepsilon)$, set

$$\hat{a} = a + \eta_1, \qquad \hat{b} = b + \eta_2, \qquad \rho = \min\left\{\max\left\{\frac{\hat{a}}{\hat{a} + \hat{b}}, 0\right\}, 1\right\},$$

and return $\tilde{m} = l + \Delta\rho$.

Original metadata comment

Taking $\alpha = \infty$, the statement reduces to checking whether the algorithm satisfies 0.8ε-DP. Under add/remove adjacency, Theorem 1 of Kulesza et al. [4] proves that the algorithm is ε-DP. Thus, the statement is not true.

Corrected proof

An ε-DP upper bound does not prove that the final clipped ratio fails to satisfy a smaller privacy parameter. We instead use a finite RDP order and a real event of the released estimate.

Without loss of generality let $l = 0, u = 1$. Write

$$U = a + \eta_1, \qquad V = b + \eta_2, \qquad \rho = \min\{\max\{U/(U + V), 0\}, 1\}.$$

Take the add/remove neighbors $D = \varnothing$ and $D' = \{1\}$, set $\varepsilon = 1$, $\alpha = 50$, and define

$$h = \frac{1}{20}, \qquad E_h = \{0 < \rho < h\}.$$

Under D, the density of ρ on $(0,1)$ is $1/2$, so

$$p = \Pr_{D}(E_h) = \frac{1}{40}.$$

Under D', $U = 1 + \eta_1$ and $V = \eta_2$. The event is equivalent to U, V having the same sign and $19|U| < |V|$. Conditioning on $U = x$ yields

$$\Pr_{D'}(E_h \mid U = x) = \frac{1}{2} e^{-19|x|}.$$

Thus

$$s = \Pr_{D'}(E_h) = \frac{1}{4} \int_{\mathbb{R}} e^{-19|x|-|x-1|}\, \mathrm{d}x = \frac{19e^{-1} - e^{-19}}{720} = 0.009707929689798066\ldots.$$

Applying (1),

$$D_{50}(M(D)\|M(D')) > \log \frac{1/40}{s} + \frac{\log(1/40)}{49} = 0.870649525365712\ldots > 0.8.$$

The claimed order-50 bound is therefore strictly violated at the ordinary privacy budget $\varepsilon = 1$.

Supplementary replace-one witness. With $D_0 = \{0\}$ and $D_1 = \{1\}$ and the same event,

$$\Pr_{D_0}(E_h) = \frac{1}{2} - \frac{19}{720}\left(19e^{-1/19} - e^{-1}\right) = 0.0340255098560050\ldots,$$

which gives $D_{50}(M(D_0)\|M(D_1)) > 1.1851747379$. This supplement is not needed for the original add/remove claim.

Verdict: No.

8 Q86: mean estimation with a rational RDP coefficient

Original problem description

For the same mean-estimation algorithm as in Q85, does the mechanism satisfy

$$\left(\alpha, \frac{(3\alpha^2 + 11\alpha + 8)\varepsilon}{5\alpha^2 + 6\alpha}\right)\text{-RDP}$$

under add/remove adjacency for every $\alpha > 1$?

Original metadata comment

Taking $\alpha = \infty$, the statement reduces to checking whether the algorithm satisfies 0.8ε-DP. Under add/remove adjacency, Theorem 1 of Kulesza et al. [4] proves that the algorithm is ε-DP. Thus, the statement is not true.

Corrected proof

The original comment contains two problems: an upper bound is not a lower-bound counterexample, and

$$\lim_{\alpha\to\infty} \frac{3\alpha^2 + 11\alpha + 8}{5\alpha^2 + 6\alpha} = \frac{3}{5} = 0.6,$$

not 0.8.

Use the same legal add/remove pair, budget, order, and public event as in Q85. The exact probabilities again give

$$D_{50}(M(D)\|M(D')) > 0.870649525365712\ldots.$$

But the coefficient claimed by Q86 at order 50 is

$$c(50) = \frac{3(50)^2 + 11(50) + 8}{5(50)^2 + 6(50)} = \frac{8058}{12800} = 0.62953125.$$

Hence $D_{50} > c(50)\varepsilon$ at $\varepsilon = 1$. This finite-order calculation is independent of the erroneous limit in the original comment.

Verdict: **No.**

9 Q88: Objective perturbation without convexity

Original problem description

Suppose that for every $z \in \mathcal{Z}$, $\ell(\cdot, z)$ is twice differentiable and the Hessian $\nabla^2\ell(w, z)$ has rank at most one for every $w \in \mathcal{W}$. Assume also that the smoothness satisfies $\beta \le \varepsilon n\lambda$. Is the following objective-perturbation algorithm (ε, δ)-DP?

Given $S = (z_1, \ldots, z_n)$, an L-Lipschitz and β-smooth loss, $\varepsilon \le 1$, $\delta \le 1/n^2$, and λ, sample

$$G \sim \mathcal{N}(0, \sigma^2 I_d), \qquad \sigma^2 = \frac{10L^2\log(1/\delta)}{\varepsilon^2},$$

and return

$$\hat{w} = \arg\min_{w\in\mathcal{W}} \left\{ \frac{1}{n}\sum_{i=1}^{n} \ell(w, z_i) + \frac{\langle G, w\rangle}{n} + \lambda\|w\|^2 \right\}.$$

Original metadata comment

The algorithm is modified from Algorithm 2 of Bassily et al. [5], removing the convexity assumption on the loss ℓ; thus it does not satisfy DP in general.

Corrected proof

The absence of convexity invalidates the cited proof route but does not by itself prove nonprivacy. We construct a nonconvex loss satisfying every remaining assumption.

Let

$$d = 1, \quad \mathcal{W} = [-1, 1], \quad n = 3, \quad \varepsilon = 1, \quad \delta = 10^{-2}, \quad L = \beta = 1, \quad \lambda = \frac{1}{3},$$

and take

$$\mathcal{Z} = \left\{0, \frac{1}{3}, \frac{1}{2}\right\}, \qquad \ell(w, c) = c(1 - w^2).$$

For every allowed w, c,

$$|\partial_w \ell(w, c)| = |2cw| \le 1, \qquad |\partial_w^2 \ell(w, c)| = 2c \le 1, \qquad \operatorname{rank}(\nabla^2 \ell) \le 1.$$

Also $\delta = 1/n^2$ and $\beta = \varepsilon n\lambda = 1$.

Take replace-one neighbors

$$S = \left(\frac{1}{3}, \frac{1}{3}, 0\right), \qquad S' = \left(\frac{1}{3}, \frac{1}{3}, \frac{1}{2}\right).$$

Here $G \sim \mathcal{N}(0, 10 \log 100)$. After removing constants that do not affect the minimizer, the two perturbed objectives are

$$F_S(w) = \frac{1}{9}w^2 + \frac{G}{3}w, \qquad F_{S'}(w) = -\frac{1}{18}w^2 + \frac{G}{3}w.$$

The first is strictly convex and has constrained minimizer

$$\hat{w}_S = \operatorname{clip}_{[-1,1]}\left(-\frac{3G}{2}\right).$$

The second is strictly concave, so every global minimizer on $[-1, 1]$ is an endpoint, independently of tie-breaking. For the public event $E = \{\hat{w} \in (-1, 1)\}$,

$$\Pr_S(E) = \Pr\{|G| < 2/3\} = \operatorname{erf}\left(\frac{2}{3\sqrt{20 \log 100}}\right) = 0.078257838375496\ldots, \qquad \Pr_{S'}(E) = 0.$$

Thus $\Pr_S(E) > e \Pr_{S'}(E) + 0.01$.

Verdict: No.

10 Q109: Confident-GNMax with the wrong RDP denominator

Original problem description

For the Confident-GNMax aggregator, let

$$\varepsilon(\lambda) = \frac{1}{\lambda} \log\left[(1 - q)A(q, \mu_2, \varepsilon_2)^{\lambda - 1} + qB(q, \mu_1, \varepsilon_1)^{\lambda - 1}\right],$$

where

$$q = \min_{i^* \in [K]} \frac{1}{2} \sum_{i \neq i^*} \operatorname{erfc}\left(\frac{n_{i^*} - n_i}{2\sigma_2}\right),$$
$$\mu_2 = \sigma_2 \sqrt{\log(1/q)}, \qquad \mu_1 = \mu_2 + 1,$$
$$\varepsilon_i = \mu_i / \sigma_2^2,$$
$$A = \frac{1 - q}{1 - (qe^{\varepsilon_2})^{(\mu_2 - 1)/\mu_2}}, \qquad B = \frac{e^{\varepsilon_1}}{q^{1/(\mu_1 - 1)}}.$$

When $q < 1$, $\mu_1 \geq \lambda$, $\mu_2 > 1$, and

$$q \leq \frac{e^{(\mu_2-1)\varepsilon_2}}{\left(\frac{\mu_1}{\mu_1-1}\frac{\mu_2}{\mu_2-1}\right)^{\mu_2}},$$

does the following algorithm satisfy data-dependent $(\lambda, \lambda/\sigma_1^2 + \varepsilon(\lambda))$-RDP?

Given a vote histogram $(n_1, \ldots, n_K)$, threshold T, and noise scales σ_1, σ_2, first test whether $\max_j n_j + \mathcal{N}(0, \sigma_1^2) \geq T$. If so, return the class maximizing $n_j + \mathcal{N}(0, \sigma_2^2)$ with independent Gaussian noises; otherwise return $\perp$.

Original metadata comment

The algorithm is the Confident-GNMax aggregator of Papernot et al. [6]. The leading factor in $\varepsilon(\lambda)$ was changed from $1/(\lambda - 1)$ to $1/\lambda$, so the original privacy proof no longer applies and the stated guarantee does not hold in general.

Corrected proof

The change in a proof formula is not itself a counterexample. We compute the mechanism's complete three-point output distribution, including $\perp$.

For two classes and a vote vector $v = (v_1, v_2)$, define

$$t(v) = \Phi\left(\frac{\max(v_1, v_2) - T}{\sigma_1}\right), \qquad \pi(v) = \Phi\left(\frac{v_1 - v_2}{\sqrt{2}\sigma_2}\right).$$

In output order $(1, 2, \perp)$, the law is

$$P_v = (t(v)\pi(v),\ t(v)(1 - \pi(v)),\ 1 - t(v)). \tag{2}$$

Take the replace-one neighboring vote histograms

$$v = (2, 0), \qquad v' = (1, 1),$$

and set

$$\sigma_2 = \frac{1}{2}, \qquad \lambda = \frac{5}{4}, \qquad \sigma_1 = 5, \qquad T = 1.$$

Then

$$q = \Phi(-2\sqrt{2}) = 0.0023388674905236\ldots,$$

$$\mu_2 = 1.230659218181112\ldots > 1, \qquad \mu_1 = 2.230659218181112\ldots > \lambda.$$

The side-condition right-hand side is

$$H_q = \frac{e^{(\mu_2-1)\varepsilon_2}}{\left(\frac{\mu_1}{\mu_1-1}\frac{\mu_2}{\mu_2-1}\right)^{\mu_2}} = 0.190700552457951\ldots > q.$$

Thus every hypothesis in the question holds. Substitution gives

$$A = 5.204443674952842\ldots, \qquad \log B = 13.845273745448896\ldots,$$

and the claimed total budget is

$$\frac{\lambda}{\sigma_1^2} + \frac{1}{\lambda}\log\left[(1-q)A^{\lambda-1} + qB^{\lambda-1}\right] = 0.416641746571765\ldots.$$

Let $t = \Phi(1/5)$. From (2), the complete laws are

$$P = (t(1-q), tq, 1-t), \qquad Q = \left(\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right).$$

Direct finite-alphabet calculation yields

$$\begin{aligned} D_{5/4}(P\|Q) &= 4\log\left[(t(1-q))^{5/4}\left(\frac{1}{4}\right)^{-1/4} + (tq)^{5/4}\left(\frac{1}{4}\right)^{-1/4} + (1-t)^{5/4}\left(\frac{1}{2}\right)^{-1/4}\right] \\ &= 0.438726899288144\ldots > 0.416641746571765\ldots. \end{aligned}$$

Verdict: No.

11 Q110: Confident-GNMax after dropping the $\mu_2 > 1$ hypothesis

Original problem description

Use the same definitions of $q, \mu_1, \mu_2, \varepsilon_1, \varepsilon_2, A, B$ and the same Confident-GNMax algorithm as in Q109, but define

$$\varepsilon(\lambda) = \frac{1}{\lambda - 1}\log\left[(1-q)A^{\lambda-1} + qB^{\lambda-1}\right].$$

Assume only

$$q \le \frac{e^{(\mu_2-1)\varepsilon_2}}{\left(\frac{\mu_1}{\mu_1-1}\frac{\mu_2}{\mu_2-1}\right)^{\mu_2}}.$$

Does the algorithm satisfy data-dependent $(\lambda, \lambda/\sigma_1^2 + \varepsilon(\lambda))$-RDP?

Original metadata comment

The algorithm is the Confident-GNMax aggregator of Papernot et al. [6]. The assumptions $q < 1$, $\mu_1 \ge \lambda$, and $\mu_2 > 1$ have been dropped, so the original privacy proof no longer applies and the stated guarantee does not hold in general.

Corrected proof

We choose parameters for which the displayed side condition is real and true, while the missing condition $\mu_2 > 1$ is exactly what fails.

Take one teacher and replace-one neighboring histograms

$$v = (1, 0), \qquad v' = (0, 1).$$

Define $\sigma_2 = s_*$ to be a root in $0.32431 < s_* < 0.32432$ of

$$s_* \sqrt{-\log \Phi\left(-\frac{1}{\sqrt{2}s_*}\right)} = \frac{2}{3}. \tag{3}$$

The left side minus $2/3$ has opposite signs at the two endpoints (approximately -3.71×10^{-6} and 3.41×10^{-6}), so such a root exists by continuity. Set

$$q = \Phi\left(-\frac{1}{\sqrt{2}s_*}\right) = 0.0146173497613541\ldots.$$

Equation (3) gives exactly

$$\mu_2 = \frac{2}{3}, \qquad \mu_1 = \frac{5}{3}.$$

The base in the side condition is negative, but its exponent is the exact rational number $2/3$. Using the real cube root,

$$\left(\frac{\mu_1}{\mu_1 - 1}\frac{\mu_2}{\mu_2 - 1}\right)^{\mu_2} = (-5)^{2/3} = 5^{2/3} > 0.$$

Writing $u = -\log q$, one has $\varepsilon_2 = 3u/2$ and therefore the side condition becomes

$$q < \frac{\sqrt{q}}{5^{2/3}} = 0.0413479818070294\ldots,$$

which is strictly true. In addition,

$$A = \frac{1-q}{1-q^{1/4}} = 1 + q^{1/4} + q^{1/2} + q^{3/4} = 1.510651128304855\ldots > 0,$$

and $B = q^{-21/4} > 0$, so every logarithm and power in the claimed budget is real.

Take $\lambda = 11/10$, $\sigma_1 = 5$, and $T = 1$. Both histograms pass the threshold with probability exactly $1/2$, and their complete laws are

$$P = \left(\frac{1-q}{2}, \frac{q}{2}, \frac{1}{2}\right), \qquad Q = \left(\frac{q}{2}, \frac{1-q}{2}, \frac{1}{2}\right).$$

Consequently,

$$D_{11/10}(P\|Q) = 10 \log\left[\frac{1}{2} + \frac{(1-q)^{11/10}q^{-1/10} + q^{11/10}(1-q)^{-1/10}}{2}\right] = 2.275068617278847\ldots.$$

The budget claimed by Q110 is

$$\frac{\lambda}{\sigma_1^2} + \frac{1}{\lambda - 1}\log\left[(1-q)A^{\lambda-1} + qB^{\lambda-1}\right] = 1.539030785591888\ldots,$$

which is strictly smaller. Notice that the dropped condition $\mu_1 \geq \lambda$ still holds here; the operative omission is $\mu_2 > 1$.

Verdict: No.

12 Q112: Propose–Test–Release with insufficient release noise

Original problem description

Given $X \in \mathcal{X}^*$ and $f : \mathcal{X}^* \to \mathbb{R}$, the question defines the proxy

$$D_\beta^f(X) = \min_{X''} \left\{ \text{Hamming}(X, X'') : \max_{X \text{ and } X' \text{ neighbors}} \|f(X) - f(X')\|_1 > \beta \right\}.$$

Does the following algorithm satisfy (ε, δ)-DP? If

$$D_\beta^f(X) + \text{Lap}(2/\varepsilon) \le \frac{2\log(1/\delta)}{\varepsilon},$$

output $\perp$; otherwise release $f(X) + \text{Lap}(\beta/\varepsilon)$.

Original metadata comment

The algorithm is modified from Propose–Test–Release [7]. The release step is ε-DP when the local sensitivity is bounded by β. A composition-style analysis gives $(3\varepsilon/2, \delta)$-DP, which is weaker than (ε, δ)-DP.

Corrected proof

The displayed proxy contains a binding typo: X'' never appears in the instability predicate. First use the standard PTR definition

$$d_\beta(X) = \min_Y \{d_H(X, Y) : \text{LS}_f(Y) > \beta\}, \qquad \text{LS}_f(Y) = \max_{Z \sim Y} |f(Y) - f(Z)|. \tag{4}$$

We show that the mechanism fails even under this corrected definition.

On binary datasets of fixed size 11, let $k(X)$ count the ones and set

$$\varepsilon = \beta = 1, \qquad \delta = e^{-5}, \qquad f(X) = \begin{cases} k(X), & 0 \le k(X) \le 10, \\ 12, & k(X) = 11. \end{cases}$$

Exactly the datasets with $k \in \{10, 11\}$ have local sensitivity exceeding one. For the replace-one neighbors

$$D = (0, \dots, 0), \qquad D' = (1, 0, \dots, 0),$$

we have

$$f(D) = 0, \quad d_1(D) = 10, \qquad f(D') = 1, \quad d_1(D') = 9.$$

The test threshold is $2\log(1/\delta) = 10$. With test noise $L \sim \text{Lap}(2)$, the release probabilities are

$$\Pr[L > 0] = \frac{1}{2}, \qquad \Pr[L > 1] = \frac{e^{-1/2}}{2}.$$

The release noise is an independent $W \sim \text{Lap}(1)$. For the public event $E = (-\infty, 0]$, excluding $\perp$,

$$\Pr_D(E) = \frac{1}{2}\Pr[W \le 0] = \frac{1}{4},$$

$$\Pr_{D'}(E) = \frac{e^{-1/2}}{2}\Pr[1 + W \le 0] = \frac{e^{-3/2}}{4}.$$

Therefore

$$\Pr_D(E) - e \Pr_{D'}(E) = \frac{1 - e^{-1/2}}{4} = 0.098367335071842\ldots > e^{-5} = \delta.$$

Literal-typo interpretation. If the unbound definition is taken literally and $\min \varnothing = +\infty$, then the proxy is zero when $\mathrm{LS}_f(X) > \beta$ and $+\infty$ otherwise. A separate two-bit example also fails: take $f(00) = 0$, $f(10) = f(01) = 1$, $f(11) = 3$, $\varepsilon = \beta = 1$, and $\delta = 0.01$. For neighbors $00, 10$, the release probabilities are 1 and $\delta/2$. On $E = (-\infty, 0]$, the probabilities are $1/2$ and $\delta/(4e)$, whereas DP would require $1/2 \le 5\delta/4 = 0.0125$.

Verdict: **No, under either the standard corrected definition or the literal typo.**

13 Q114: A PTR variant calibrated to local sensitivity

Original problem description

For $f : \mathcal{X} \to \mathbb{R}$, define

$$\Delta_{\mathrm{LS}}(X) = \max_{X' \sim X} |f(X) - f(X')|.$$

Suppose an oracle $\mathcal{T} : \mathcal{X} \to \{0, 1\}$ is (ε, δ)-DP and, whenever $\Delta_{\mathrm{LS}}(X) \ge \Delta/\varepsilon$, satisfies $\Pr[\mathcal{T}(X) = 1] \le \delta$. Does the following algorithm satisfy $(2\varepsilon, 2\delta)$-DP?

If $\mathcal{T}(X) = 0$, output $\perp$. Otherwise release

$$\theta = f(X) + \mathrm{Laplace}\left(\frac{\Delta_{\mathrm{LS}}(X)}{\varepsilon}\right).$$

Original metadata comment

This is modified from Algorithm 2 of Redberg, Zhu, and Wang [8]. The release step is not ε-DP because its noise is calibrated to local sensitivity rather than to the proposed scale Δ. Thus the full procedure is not $(2\varepsilon, 2\delta)$-DP in general.

Corrected proof

Work on the fixed-size binary domain $\{0, 1\}^2$. Set

$$\varepsilon = 1, \qquad \delta = 0.01, \qquad \Delta = 5,$$

and define

$$f(00) = 0, \qquad f(10) = f(01) = 1, \qquad f(11) = 5.$$

The complete local-sensitivity table is

X	00	10	01	11
$f(X)$	0	1	1	5
$\Delta_{\mathrm{LS}}(X)$	1	4	4	4

Take the deterministic oracle $\mathcal{T}(X) \equiv 1$. It is $(0, 0)$-DP and hence satisfies the required $(1, 0.01)$-DP

condition. The false-positive premise also holds, because $\Delta_{\mathrm{LS}}(X) \le 4 < \Delta/\varepsilon = 5$ for every X; its antecedent is never true.

For the neighbors $D = 00$ and $D' = 10$, the algorithm always releases, with

$$M(D) \sim \mathrm{Lap}(1), \qquad M(D') \sim 1 + \mathrm{Lap}(4).$$

On the common public event $E = [5, \infty)$,

$$\Pr_{D}(E) = \frac{e^{-5}}{2}, \qquad \Pr_{D'}(E) = \frac{e^{-1}}{2}.$$

In the direction $D' \to D$,

$$\Pr_{D'}(E) - e^{2} \Pr_{D}(E) = \frac{e^{-1} - e^{-3}}{2} = 0.159046186401789\ldots > 0.02 = 2\delta.$$

All oracle premises hold; the failure comes from choosing a data-dependent release scale equal to local sensitivity.

Verdict: No.

14 Q116: The single-quantile mechanism

Original problem description

Does the following quantile mechanism satisfy $(\varepsilon/2)$-DP?

Given $q \in [0, 1]$, $X = (x_1, \ldots, x_n) \in (a, b)^n$, and ε, define

$$u(X, w) = -\left|\#\{x \in X : x < w\} - \lfloor qn \rfloor\right|.$$

Partition (a, b) at the sorted data into intervals $I_k = [x_{k-1}, x_k]$, with $x_0 = a$ and $x_{n+1} = b$. Sample I_k with probability proportional to

$$\exp\left(\frac{\varepsilon u(X, I_k)}{2}\right) |I_k|,$$

then output v uniformly from the selected interval.

Original metadata comment

The algorithm is the DP single-quantile mechanism of Appendix A in Kaplan, Schnapp, and Stemmer [9]. It satisfies ε-DP by their Lemma A.1.

Corrected proof

An ε-DP upper bound does not disprove $(\varepsilon/2)$-DP. The public output is continuous, and the data-dependent interval length cancels with the conditional uniform density. Thus the actual density

is

$$p_X(w) = \frac{e^{\varepsilon u(X,w)/2}}{Z_X}, \qquad Z_X = \int_a^b e^{\varepsilon u(X,v)/2}\,\mathrm{d}v. \tag{5}$$

Let

$$(a,b) = (0,1), \qquad n = 2, \qquad q = \frac{1}{2}, \qquad r = e^{-\varepsilon/2} \in (0,1),$$

and take replace-one neighbors and one common output event

$$X = \left(\frac{1}{8}, \frac{1}{4}\right), \qquad X' = \left(\frac{1}{4}, \frac{7}{8}\right), \qquad E = \left(\frac{1}{8}, \frac{1}{4}\right).$$

The optimal-utility region has weight one and every other region has weight r. Hence

$$Z_X = \frac{1+7r}{8}, \qquad Z_{X'} = \frac{5+3r}{8}.$$

Using (5),

$$\Pr_X(E) = \frac{1}{1+7r}, \qquad \Pr_{X'}(E) = \frac{r}{5+3r}.$$

Therefore

$$\frac{\Pr_X(E)}{\Pr_{X'}(E)} = e^{\varepsilon/2}\frac{5+3r}{1+7r} > e^{\varepsilon/2}, \qquad 0 < r < 1.$$

This strict violation holds for every $\varepsilon > 0$.

The result is consistent with the mechanism's true ε-DP guarantee. For replace-one neighbors X, X' and every common output w, $|u(X,w) - u(X',w)| \le 1$; the unnormalized-density ratio and the normalizer ratio can each contribute $e^{\varepsilon/2}$, for a total of e^{ε}.

Verdict: **No.**

References

[1] R. McKenna and D. Sheldon. Permute-and-Flip: A New Mechanism for Differentially Private Selection. *NeurIPS*, 2020.

[2] J. Dong, D. Durfee, and R. Rogers. Optimal Differential Privacy Composition for Exponential Mechanisms. *ICML*, 2020.

[3] O. Sheffet. Differentially Private Ordinary Least Squares. *ICML*, 2017.

[4] A. Kulesza et al. Mean Estimation in the Add–Remove Model of Differential Privacy. *ICML*, 2024.

[5] R. Bassily, V. Feldman, K. Talwar, and A. Thakurta. Private Stochastic Convex Optimization with Optimal Rates.

[6] N. Papernot et al. Scalable Private Learning with PATE. 2018.

[7] C. Dwork and J. Lei. Differential Privacy and Robust Statistics. 2009.

[8] R. Redberg, Y.-X. Zhu, and Y.-X. Wang. Generalized PTR: User-Friendly Recipes for Data-Adaptive Algorithms with Differential Privacy. 2023.

[9] H. Kaplan, S. Schnapp, and U. Stemmer. Differentially Private Approximate Quantiles. *ICML*, 2022, Appendix A.